Covariate-Guided Clusterwise Linear Regression for Generalization to Unseen Data

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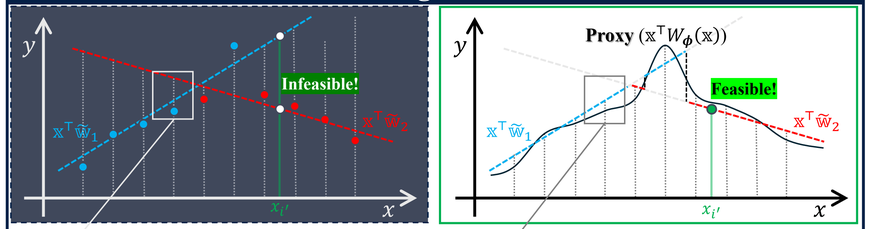
paper github

Test-time Assignment Problem



Response-aware Assignment
$z_i = \mathrm{argmin}_k \left(y_i - \mathbb{x}_i^\top \tilde{\mathbb{w}}_k\right)^2$

Proxy-based Assignment
$z_i = \mathrm{argmin}_k \left(\mathbb{x}_i^\top W_\phi(\mathbb{x}_i) - \mathbb{x}_i^\top \tilde{\mathbb{w}}_k\right)^2$

CG-CLR Architecture

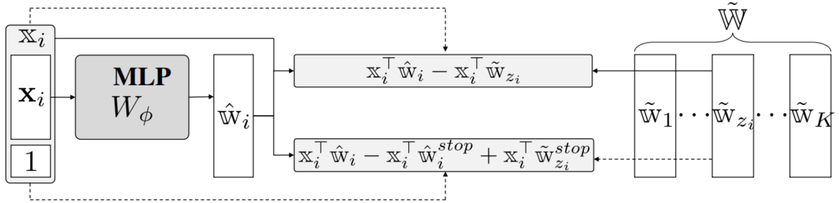


Figure 1: **CG-CLR architecture.** The proxy network W_ϕ maps a covariate $\boldsymbol{x}_i \in \mathbb{R}^p$ to a proxy output vector $\hat{\mathbb{w}}_i \in \mathbb{R}^{p+1}$. The vector-quantizer compares the proxy prediction $\mathbb{x}_i^\top \hat{\mathbb{w}}_i$ with the K codebook vector predictions $\{\mathbb{x}_i^\top \tilde{\mathbb{w}}_j\}_{j=1}^K$ and outputs the index z_i. The selected codebook vector $\tilde{\mathbb{w}}_{z_i}$ produces the final prediction $\mathbb{x}_i^\top \tilde{\mathbb{w}}_{z_i}$. During back-propagation, the stop-gradient copies (dashed arrows) prevent the proxy path from negatively affecting the current clusterwise fit.

Objective

Alignment loss

$$\mathbb{E}_{\mathbb{x}_i \sim \boldsymbol{D}}\left[\left(\mathbb{x}_i^\top W_\phi(\mathbb{x}_i) - \mathbb{x}_i^\top \tilde{\mathbb{w}}_{z_i}\right)^2\right]$$

Alignment loss pushes the proxy prediction toward their codebook counterparts so that future assignments are improved.

Prediction-fit loss

$$\mathbb{E}_{(\mathbb{x}_i, y_i) \sim \boldsymbol{D}}\left[\left(\left(y_i - \mathbb{x}_i^\top W_\phi(\mathbb{x}_i)\right) + \mathbb{x}_i^\top\left(W_\phi(\mathbb{x}_i) - \tilde{\mathbb{w}}_{z_i}\right)^{stop}\right)^2\right]$$

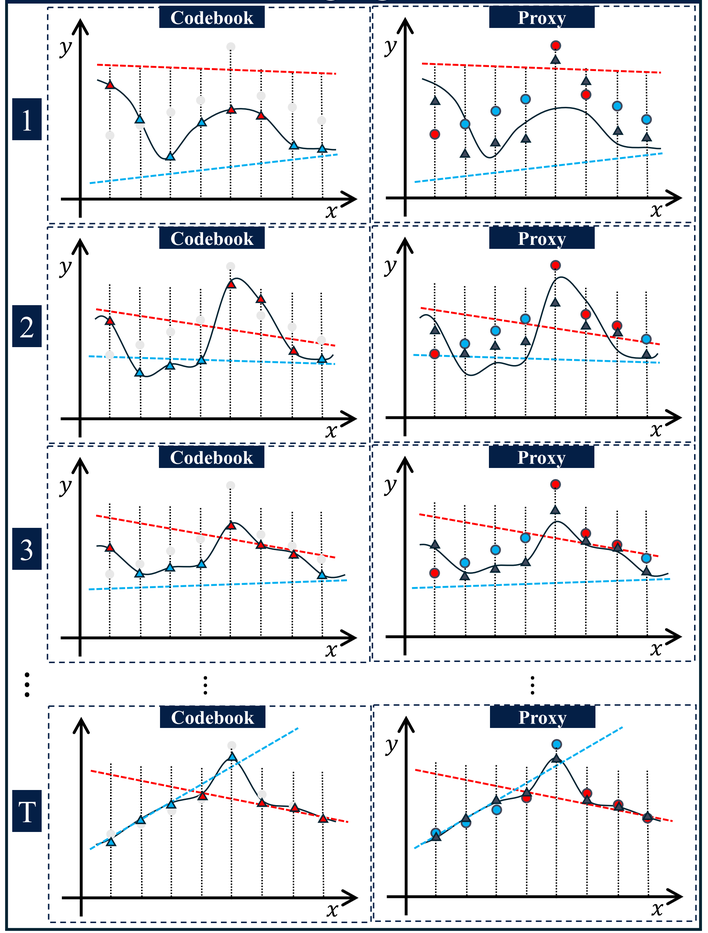
Prediction-fit loss keeps each regressor faithful to its currently assigned samples.

Composite Objective. $V_\lambda(\boldsymbol{\phi}, \tilde{\mathbb{W}}) := \mathbb{E}_{(\mathbb{x}_i, y_i) \sim \boldsymbol{D}}\left[\left(\mathbb{x}_i^\top W_\phi(\mathbb{x}_i) - \mathbb{x}_i^\top \tilde{\mathbb{w}}_{z_i}\right)^2 + \left(\left(y_i - \mathbb{x}_i^\top W_\phi(\mathbb{x}_i)\right) + \mathbb{x}_i^\top\left(W_\phi(\mathbb{x}_i) - \tilde{\mathbb{w}}_{z_i}\right)^{stop}\right)^2\right]$

Codebook Update. $\tilde{\mathbb{w}}_j^{(t+1)} \leftarrow \tilde{\mathbb{w}}_j^{(t)} + 2\eta(1+\lambda)\, \mathbb{E}_{\mathbb{x}_i \sim \boldsymbol{D} | z_i = j}\left[\mathbb{x}_i \mathbb{x}_i^\top \left(W_{\phi^{(t)}}(\mathbb{x}_i) - \tilde{\mathbb{w}}_j^{(t)}\right)\right]$

Proxy Update. $\boldsymbol{\phi}^{(t+1)} \leftarrow \boldsymbol{\phi}^{(t)} + 2\eta(1+\lambda)\, \mathbb{E}_{(\mathbb{x}_i, y_i) \sim \boldsymbol{D}}\left[\left(\frac{y_i + \lambda \mathbb{x}_i^\top \tilde{\mathbb{w}}_{z_i}^{(t)}}{1+\lambda} - \mathbb{x}_i^\top W_{\phi^{(t)}}(\mathbb{x}_i)\right) \nabla_\phi \left(W_{\phi^{(t)}}(\mathbb{x}_i)\right)^\top \mathbb{x}_i\right]$

Training Algorithm



Theoretical Analysis

Assumption 1. *For each fixed $\boldsymbol{x}$, the function $W_\phi(\boldsymbol{x})$ is L_x-Lipschitz continuous and its Jacobian satisfies $\|\nabla_\phi W_\phi(\boldsymbol{x})\, \boldsymbol{v}\| \geq m_x \|\boldsymbol{v}\| > 0$ for all $\boldsymbol{v}$.*

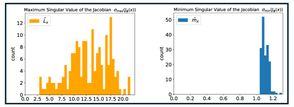


Figure 2: Empirical Jacobian spectrum validating Assumption 1: minimum(left) and maximum(right)

Assumption 2. *For a fixed cluster assignment z, the alignment loss R^a satisfies $\mu \boldsymbol{I} \preccurlyeq \nabla^2_{\tilde{\mathbb{W}}} R^a \preccurlyeq L\boldsymbol{I}$, for constants $\mu, L > 0$.*

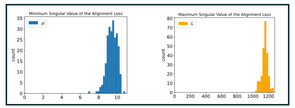


Figure 3: Empirical alignment loss Hessian spectrum validating Assumption 2: minimum(left) and maximum(right) eigenvalues.

Assumption 3. *For the optimal $\tilde{\mathbb{W}}^*$, $\min_{j \neq k} \min_i \left|\mathbb{x}_i^\top \tilde{\mathbb{w}}_j^* - \mathbb{x}_i^\top \tilde{\mathbb{w}}_k^*\right| \geq \Delta > 0$.*

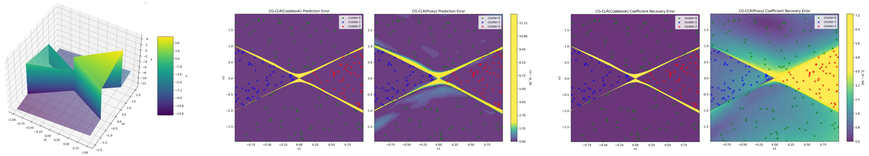
Assumption 4. *The proxy network is at least as expressive as the class of CLR.*

Theorem (Linear convergence). *Let Assumptions 1—4 hold. Define*

$$\mu_V := \min\{2m_x^2 \mathbb{x}_{\min}^2, (1+\lambda)\mu\}, \qquad L_V := \max\{(1+\lambda)L, (4+2\lambda)L_x^2 \mathbb{x}_{\max}^2\}, \qquad q := \frac{L_V - \mu_V}{L_V + \mu_V}.$$

Gradient descent with the optimal step-size $\eta_* = 2/(\mu_V + L_V)$ ensures that the stacked parameter vector $\boldsymbol{\theta} = (\boldsymbol{\phi}, \tilde{\mathbb{W}})$ converges linearly at the rate q, while Lyapunov gap contracts at the rate q^2.

Simulation Study



(a) True Function **(b) Prediction Error (Codebook | Proxy)** **(c) Recovery Error (Codebook | Proxy)**

Figure 2: Synthetic piecewise linear function, prediction error ($|y_i - \mathbb{x}_i^\top \tilde{\mathbb{w}}_{z_i}|, |y_i - \mathbb{x}_i^\top W_\phi(\mathbb{x}_i)|$), and recovery error ($|\mathbb{w}_i - \tilde{\mathbb{w}}_{z_i}|, |\mathbb{w}_i - W_\phi(\mathbb{x}_i)|$). Error maps compare codebook-based (CLR) predictions and proxy-based predictions.

Real-world Benchmarks

Table 1: Test RMSE (95% confidence intervals) across the seven datasets with $K = \lfloor N/(10p+10) \rfloor$. We report results for representative CLR-based approaches grouped by their assignment mechanisms: (i) response-aware assignment methods (MLR*, EM-MLR*) that assign clusters via nearest neighbors; (ii) tree-based partitioning methods (CART, PILOT, LDT) that learn axis-aligned splits and fit constant or linear experts; (iii) neural gating method (S-IME$_d$) that learn a gating network to select linear experts.

Model	CONDUCT	HOUSING	BIKE	ELECTRICAL	PLANT	WINE	CONCRETE
MLR*	[33.77, 33.95]	[1.159, 1.169]	[177.57, 178.91]	[0.035, 0.035]	[13.717, 13.877]	[0.827, 0.843]	[15.037, 15.449]
EM-MLR*	[18.75, 19.11]	[0.748, 0.768]	[102.28, 103.62]	[0.022, 0.022]	[5.208, 5.406]	[0.726, 0.747]	[10.404, 10.938]
CART	[16.56, 16.86]	[0.645, 0.659]	[120.34, 122.30]	[0.021, 0.021]	[4.217, 4.305]	[0.734, 0.756]	[9.316, 9.822]
PILOT	[15.31, 15.74]	[0.816, 0.826]	[165.26, 166.48]	[0.033, 0.033]	[4.711, 4.837]	[0.746, 0.766]	[14.236, 14.824]
LDT	[13.36, 15.62]	[0.603, 0.743]	[59.44, 60.02]	[0.015, 0.015]	[4.076, 4.162]	[0.698, 0.718]	[6.110, 6.418]
S-IME$_d$	[12.68, 13.04]	[0.560, 0.570]	[56.13, 58.01]	[0.010, 0.010]	[4.183, 4.259]	[0.693, 0.715]	[8.853, 9.511]
CG-CLR (CODEBOOK)	**[10.50, 10.62]**	**[0.485, 0.497]**	**[40.77, 41.71]**	**[0.006, 0.006]**	**[3.573, 3.675]**	**[0.652, 0.676]**	**[5.193, 5.537]**